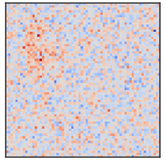
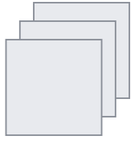
## Inputs



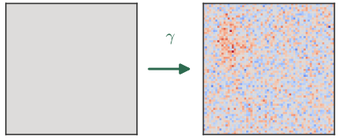
$\mathbf{x}_\star$
target instance

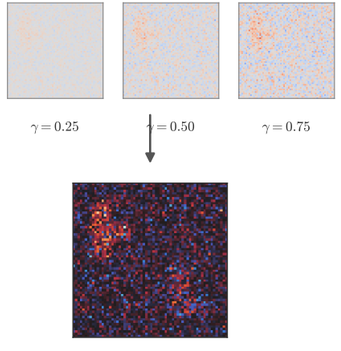
$f$ differentiable classifier



$\mathcal{D}_{\text{train}}$
training samples

## 1 Deterministic IG



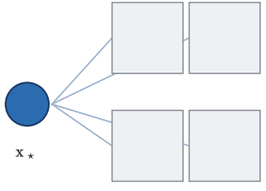




$\mathbf{w}^{\text{PG}} = \hat{\mathbf{s}}^{\text{IG}}(\mathbf{x}_\star)$
pixel-level IG attribution

deterministic: fixed $(f, \mathbf{x}', M)$
no random masks, no surrogate fit
(M-step quadrature error only)

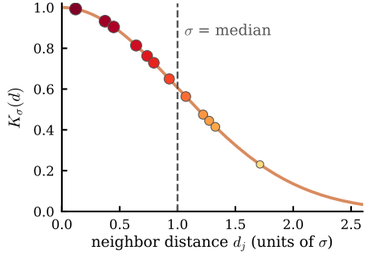
## 2 Neighborhood

$\mathcal{N}_K(\mathbf{x}_\star)$




$K$ nearest training samples
under the Euclidean metric;
retrieved from $\mathcal{D}_{\text{train}}$,
not synthesized perturbations

## 3 Gaussian locality






weighted, centered neighborhood $\widetilde{\mathbf{X}}_K$

$\pi_j = K_\sigma(d_j) \,/\, \sum_{\ell=1}^{K} K_\sigma(d_\ell)$
adaptive bandwidth $\sigma = \text{median}\, d_j$;
no bandwidth hyper-parameter;
not a retraining step

## 4 Pattern conversion

$$\mathbf{a} = \frac{\mathrm{Cov}_{\Pi^K}[\mathbf{h}_\star(\mathbf{x}),\, \hat{y}]}{\mathrm{Var}_{\Pi^K}[\hat{y}] + \lambda}$$

$$\mathbf{a} = \frac{\mathbf{C}_K\, \mathbf{w}^{\text{PG}}}{(\mathbf{w}^{\text{PG}})^\top\, \mathbf{C}_K\, \mathbf{w}^{\text{PG}} + \lambda}$$

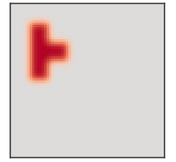
$\mathbf{w}^{\text{PG}}$ → $\mathbf{C}_K$ never materialized → $\mathbf{a}$

two-pass product:
$\widetilde{\mathbf{X}}_K^\top(\boldsymbol{\pi} \odot (\widetilde{\mathbf{X}}_K\, \mathbf{w}^{\text{PG}}))$

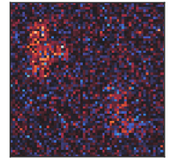
$\mathcal{O}(KD)$ time and space;
no $D \times D$ matrix; $\lambda = 10^{-5}$

## Comparison

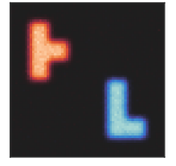
same instance, same scale



ground-truth signal



raw Integrated Gradients



PatternGrad